## Roots of a Polynomial Equation

In general, if $p(a) = 0$ we say that $a$ is zero of polynomial $p(x)$ or $a$ is the root of polynomial equation $p(x) = 0$

**Example 3.7** If $f(x) = x^2 - 4x + 3$, then find the values of $f(1)$, $f(-1)$, $f(2)$, $f(3)$. Also find the zeros of the polynomial $f(x)$.

***Solution***

$$f(x) = x^2 - 4x + 3$$

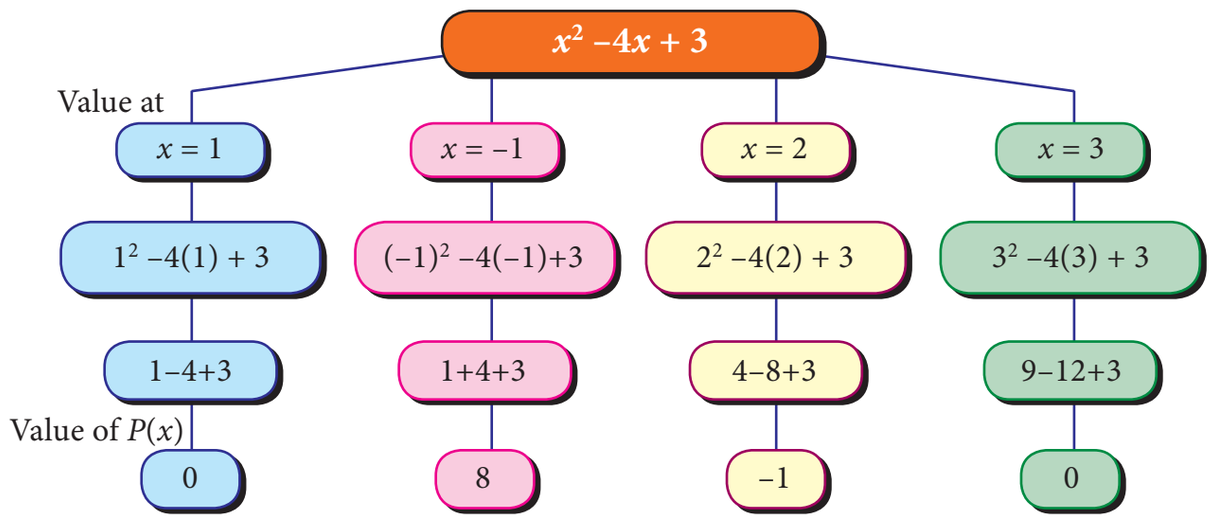



Fig. 3.9

Since the value of the polynomial $f(x)$ at $x = 1$ and $x = 3$ is zero, as the zeros of polynomial $f(x)$ are 1 and 3.

**Example 3.8** Find the Zeros of the following polynomials.

(i) $f(x) = 2x + 1$ (ii) $f(x) = 3x - 5$

***Solution***

(i) Given that $f(x) = 2x + 1 = 2\left(x + \frac{1}{2}\right)$

$$= 2\left(x - \left(-\frac{1}{2}\right)\right)$$

$$f\left(-\frac{1}{2}\right) = 2\left[-\frac{1}{2} - \left(-\frac{1}{2}\right)\right] = 2(0) = 0$$

Since $f\left(-\frac{1}{2}\right) = 0$, $x = -\frac{1}{2}$ is the zero of $f(x)$

(ii) Given that $f(x) = 3x - 5 = 3\left(x - \frac{5}{3}\right)$

$$f\left(\frac{5}{3}\right) = 3\left(\frac{5}{3} - \frac{5}{3}\right) = 3(0) = 0$$

Since $f\left(\frac{5}{3}\right) = 0$, $x = \frac{5}{3}$ is the zero of $f(x)$



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